addac112 CD SKIP

This is the stock addac112 firmware (copied at the commit in `UPSTREAM`) with a second engine for the left knob section: the CD-skip engine in `../engine/`. The looper, banks, SD card, presets, loop pitch, sync and REC prob/delay are the same in both modes.

- **GRANULAR** is the stock 112: same grains, menus and presets, and 360 s of loop memory.
- **CD SKIP** turns the left section into a skipping-CD effect. Skips replay the **live input** (a window of up to 20 s) or the **selected loop**, optionally locked to its slices. It has ~284 s of loop memory, because the live window takes the rest.

```
make                # build/CDSKIPPER_<version>.bin
make program-app    # bootloader in DFU mode
```

It builds against addac112's libdaisy / DaisySP / CMSIS (`make ADDAC112_DIR=...` to override). libdaisy is linked at `-O2` (`build_libdaisy.sh` builds it into `libdaisy_O2/`, and `make` runs it when it's missing; the vendored build is `-O0`). The skip engine (`skip_bridge.cpp`) and the looper (`granular.cpp`, `loop.cpp`) are built with `-O2`, and the looper's per-sample code and libm run from ITCM (`STM32H750IB_sram_itcm.lds`). The audio callback is timed in CPU cycles, because `hw.GetUs()` wraps every ~17.9 s.

Switching engines

Each engine is its own firmware. This one always runs CD SKIP. Plain granular is the stock addac112 firmware.

1. Put the firmware `.bin` files (112, CD SKIP, FRIPPER, ...) in a folder named `ENGINES` on the SD card.
2. Go to **Menu** → **ENGINE** (the last item of the main menu). It lists the `.bin` files in `ENGINES`.
3. Choose one and confirm **FLASH & RESTART? YES**.

- The module programs the chosen firmware and restarts into it. The `.bin` stays in `ENGINES`, so you can switch back the same way. The bank selection and the calibration are kept.
- `.bin` files in the SD root are still installed by the bootloader at power-up, as before. The bootloader ignores the `ENGINES` folder.
- The file is checked before anything is erased. If the SD card fails during the write, the module stays on and asks you to flash again. Do this before powering off. If that is not possible, put a `.bin` in the SD root and power-cycle.
- The old `engine_mode=` key in `GLOBAL.CFG` is ignored.

Presets store both engines' values (the stock grain keys plus `skip_*` keys). The stock 112 firmware ignores the `skip_*` keys, but if it saves a preset, those keys are dropped from that preset.

CD SKIP controls (dual-label overlay)



The engine's labels are in orange; the rest of the panel works as on the stock 112.

Row	Col 1	Switch (FREE / ÷8 / ÷4)	Col 2	Col 3
1	POSITION / **WINDOW**	START GRID	POS DEV / **DRIFT**	GRAINS ACTIVE / **LAYERS** (jack in trigger mode: **SKIP TRIG**)
2	PLAY LENGTH / **SIZE**	SIZE GRID	LENGTH DEV / **SPREAD**	ATTACK / **SMOOTH**
3	GRAIN DELAY / **DENSITY**	SKIP CLOCK	DELAY DEV / **ERROR**	DECAY / **TONE**
4	DIRECTION / **HOLD**	REPEAT / **SOURCE** (LIVE / LOOP)	PROB/REPEAT / **DROPOUT**	GRAIN PITCH / **PITCH** (+1V/oct)
5	VOLUME MIN / **DUCK**	–	VOLUME DEV / **WASH**	PANNING / **WIDTH**

Grids are ÷8 / ÷4 slices of the selected loop:

- **START GRID**: skips start on slices.
- **SIZE GRID**: the stuck-loop length is a slice × 2^k, and SPREAD changes k.
- **SKIP CLOCK**: one skip chance per slice, with probability = DENSITY.

Other panel controls in CD SKIP mode:

- **GRAINS VOLUME** = skip level.
- **GRAINS FEEDBACK** = skips re-recorded into the loop.
- **QUANTIZE** sets the scale for PITCH and DETUNE.
- The skips duck the loop playback and the dry input (DUCK; DUCK TARGET in the menu).

Display (CD SKIP):

- Turning an overlay knob or switch shows its name and value for 1.5 s, with `PICKUP` until it reaches a preset's value.
- The main screen shows one bar per skip, the window, a duck meter (left) and WIDTH (right).

- Top bar: `L<layers>` / `T<n>`. Bottom bar: `IN` / `LP` source, the SMOOTH envelope, and G = skip transposition.
- **SKIP SETTINGS** pages replace GRAINS SETTINGS, last after the OTHER SETTINGS pages (also reachable from Menu → Options → 10.SKIP OPTIONS, the last menu pages):
  - SKIP TRIGGER, DETUNE, REVERB DECAY, DUCK TARGET / ATTACK / RELEASE, FOLLOW LOOP PITCH, LAYERS FULL;
  - DUCK KEY / THRESHOLD / RATIO, CD SECTORS, FREEZE LIVE.
- Encoder + STOP (special function) clears the buffer: the skips and the wash fade out, and the history starts empty.

Memory

	GRANULAR	CD SKIP
Loop memory	360 s (stock)	~284 s
Live history (2 × 2^21 floats) + reverb lines	–	top of the loop array in SDRAM
Skip engine object (8 KB)	–	heap (RAM_D2), created at boot
Grain voice reserve (21 KB heap)	yes	no

DTCM (shared with the stack) is at the stock level (88.6 %). In CD SKIP mode at 96 kHz the live window is ~16 s.

Web editor

`web/` has a Skip section next to Grains, and ENGINE on the card tab. Run its tests with `for t in web/test/*.test.js; do node "$t"; done` (the sample files are read from `../../addac112/support`).

Not done yet

- Not yet run on hardware.
- Overlay artwork: needs the panel dimensions / knob positions.